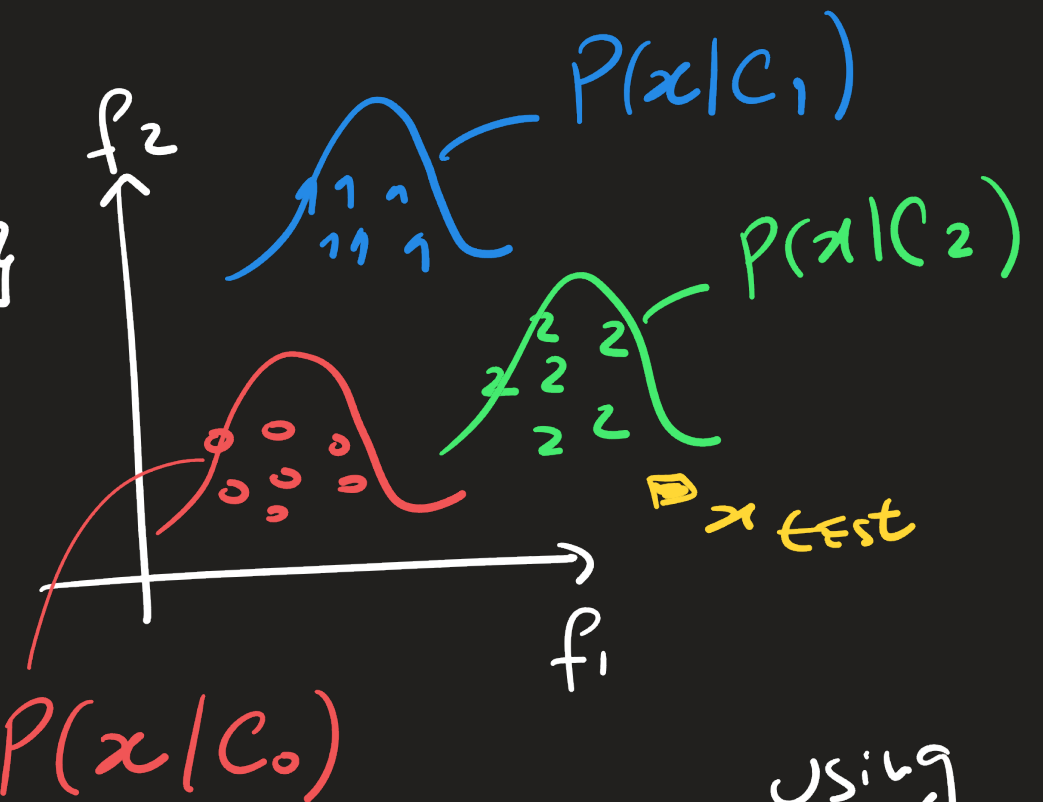


Naive Bayes classifier

$$\{(x_i, t_i)\}_{i=1}^N, \quad t_i \in \{0, 1, 2\}$$
$$x_i \in \mathbb{R}^2$$

generative classifier



① $P(x|C_j)$ $\equiv$ data like likelihood
for class C_j

using
MLE
or
MAP

Note, each $P(x|C_j)$ requires estimation
of its parameters. If they are all
assumed to be Gaussian (bivariate),

then the parameters $\{\mu_j, \Sigma_j\}_{j=0}^2$.

② Estimate prior for each class, $P(C_j)$

$$P(C_j) = \frac{\# \text{ samples in } C_j}{\# \text{ total samples}}$$

Note: this would NOT be the preferred choice when classes are *imbalanced*.

③ Estimate the posterior for each class:

$$\underline{P(C_j | x)} = \frac{P(x | C_j) \cdot P(C_j)}{\sum_{k=0}^2 P(x | C_k) \cdot P(C_k)} \in [0, 1]$$

$$\sum_{j=0}^2 P(C_j | x) = 1$$

Predictions:

x_{test}

$$P(C_0 | x_{\text{test}}), P(C_1 | x_{\text{test}}), P(C_2 | x_{\text{test}})$$

select class for
each if
posterior is largest.

In general,

$$x \in C_K \text{ if } K = \underset{j}{\operatorname{arg\,max}} P(C_j | x)$$

Finding the expression for the decision surface:

$$\frac{P(C_1|x)}{P(C_2|x)} > 1, \quad x \in C_1$$

$$\frac{\frac{P(x|C_1) \cdot P(C_1)}{P(x|C_1) \cdot P(C_1) + P(x|C_2) \cdot P(C_2)}}{\frac{P(x|C_2) \cdot P(C_2)}{P(x|C_1) \cdot P(C_1) + P(x|C_2) \cdot P(C_2)}} > 1$$

$$\ln \left(\frac{P(x|C_1) \cdot P(C_1)}{P(x|C_2) \cdot P(C_2)} \right) > \ln(1) = 0$$

$$\underbrace{\ln(P(x|C_1) \cdot P(C_1))}_{g_1(x)} - \underbrace{\ln(P(x|C_2) \cdot P(C_2))}_{g_2(x)} > 0$$

$$\underbrace{\hspace{10em}}_{g(x)}$$